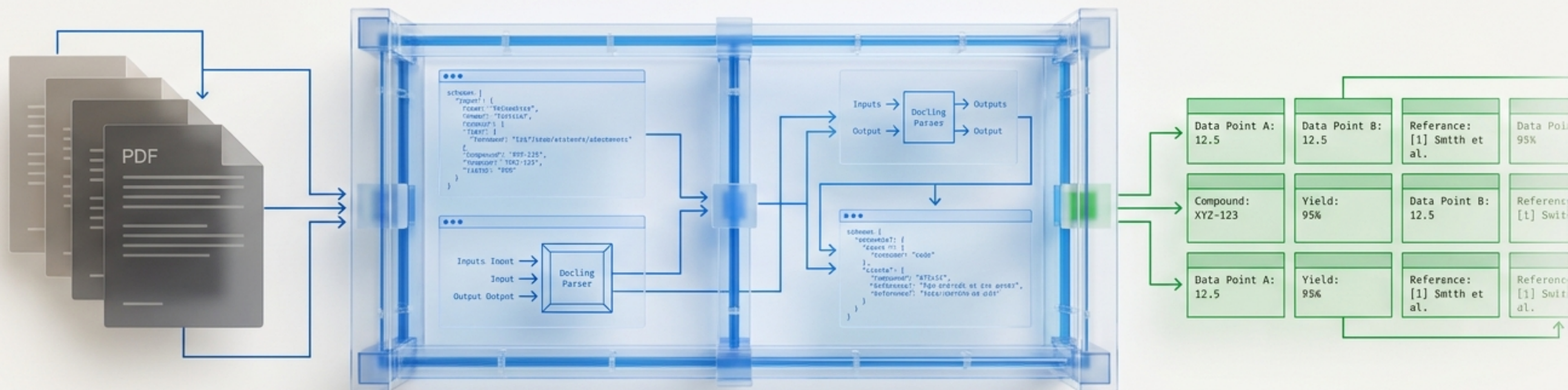


papers-to-table

Local-first, AI-assisted data extraction from scientific literature.

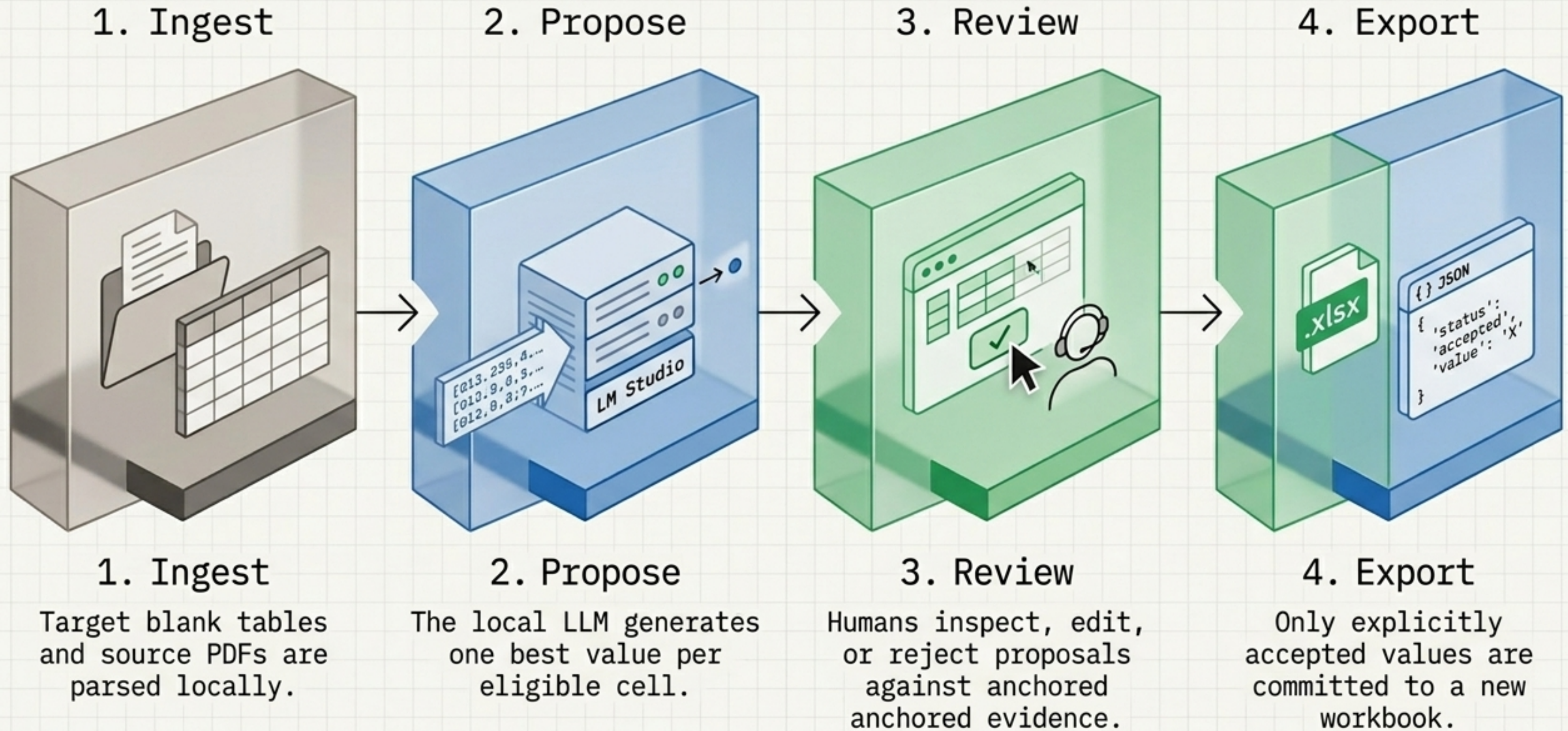


mode: normal_extraction | locality: local | parser: docling

Constrained Extraction Over Autonomous Editing

✗ AUTONOMOUS SPREADSHEET POPULATION	✓ AI-ASSISTED HUMAN REVIEW WORKFLOW
Autonomous spreadsheet population	AI-assisted human review workflow
Broad, open-ended chatbot prompts	Strict schema-driven target cell extraction
Unauditable black box answers	Visibly anchored source text and structured evidence
Direct, unreviewed writes to the active workbook	Explicit, human-approved exports to a brand-new XLSX file

From Raw Literature to Audited Tables



Three-Panel Diagnostic Review Workspace

Diagnostics Drawer:
Warning count,
unmatched PDFs,
duplicates.

**Actionable Proposal
Queue:**
Grouped by
paper/column.
Filters out
unreviewable noise.

[Paper_A.pdf / Column_B]

[Paper_C.pdf / Column_A]

[Paper_A.pdf / Column_A]

[Paper_A.pdf / Column_B]

[Paper_C.pdf / Column_B]

[Paper_A.pdf / Column_A]

[Paper_C.pdf / Column_A]

FIELD: Extract Value X

PROPOSED:

123.45

ORDERED EVIDENCE:

[LINK 1: pg. 2]

[LINK 2: pg. 4]

Accept Edit

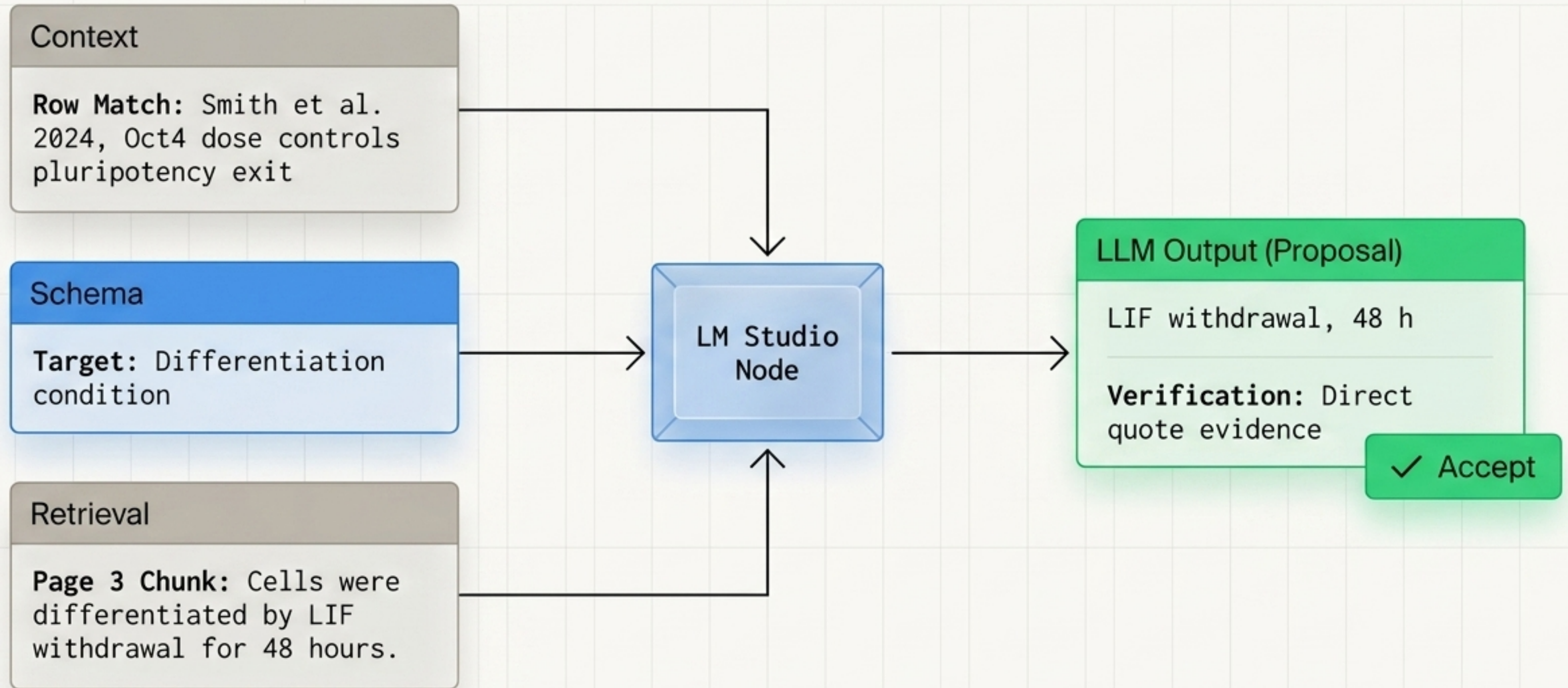
Confirm No Data Reject

Evidence & PDF Viewer:
Displays source
document with exact
highlighted anchors.

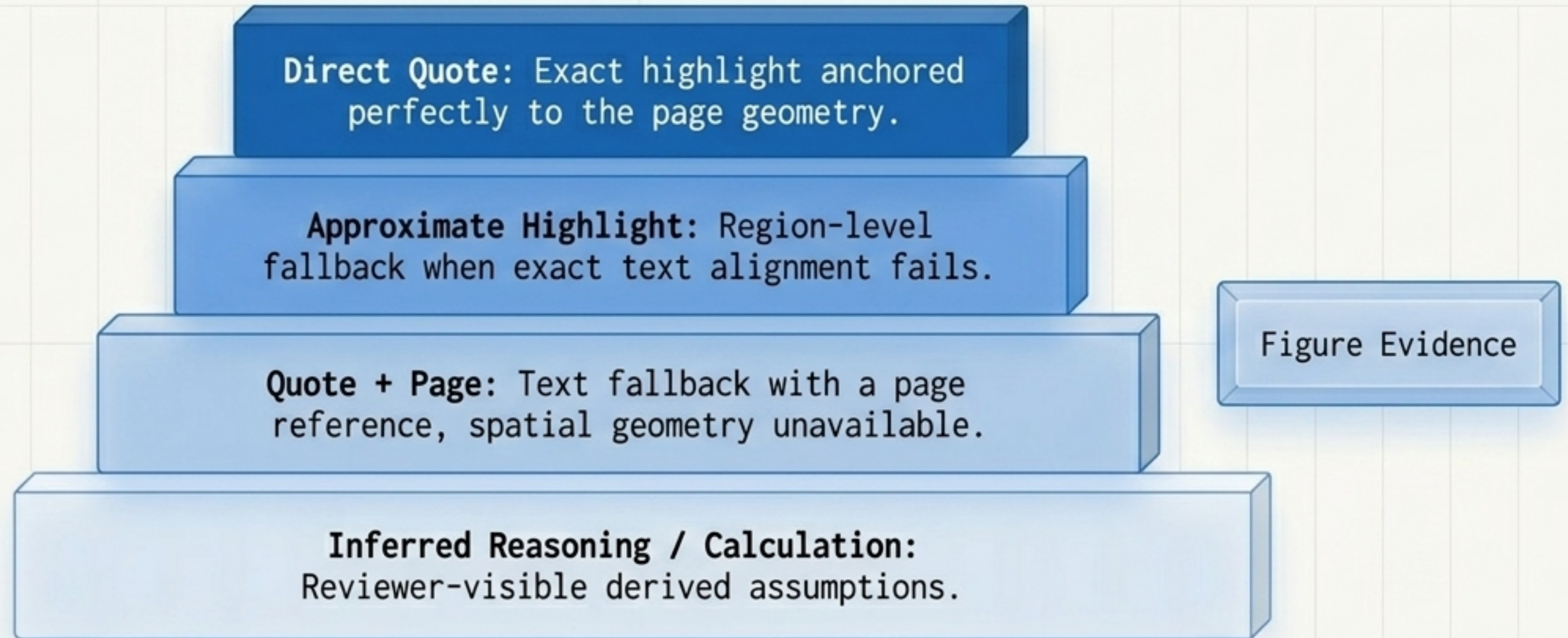
Scrolls independently.

Proposal Detail:
Selected field, proposed value,
ordered evidence links, and human decisions
(Accept / Edit / Confirm No Data / Reject).

Anatomy of a Single Cell Extraction

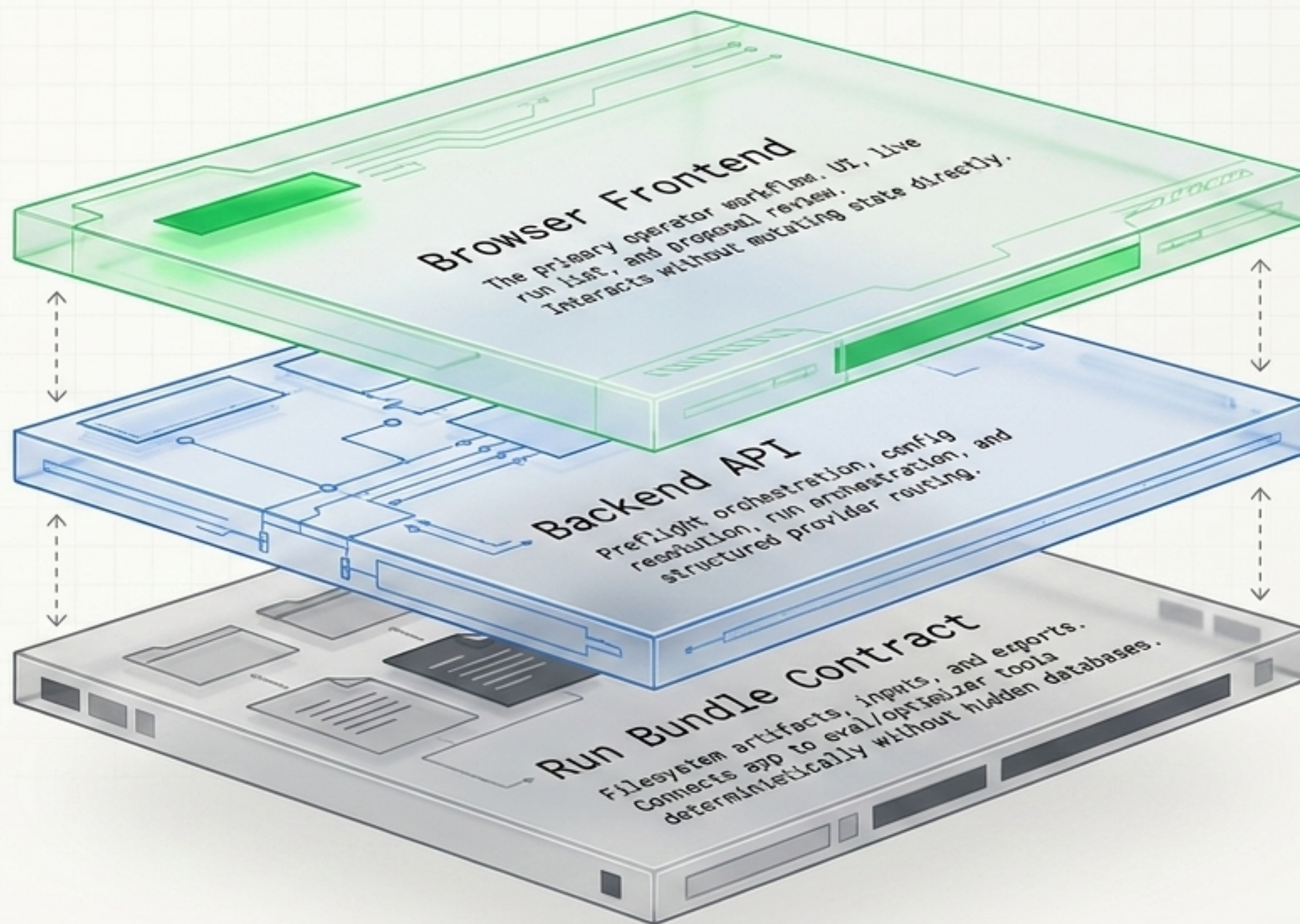


Evidence Semantics and Support Authority

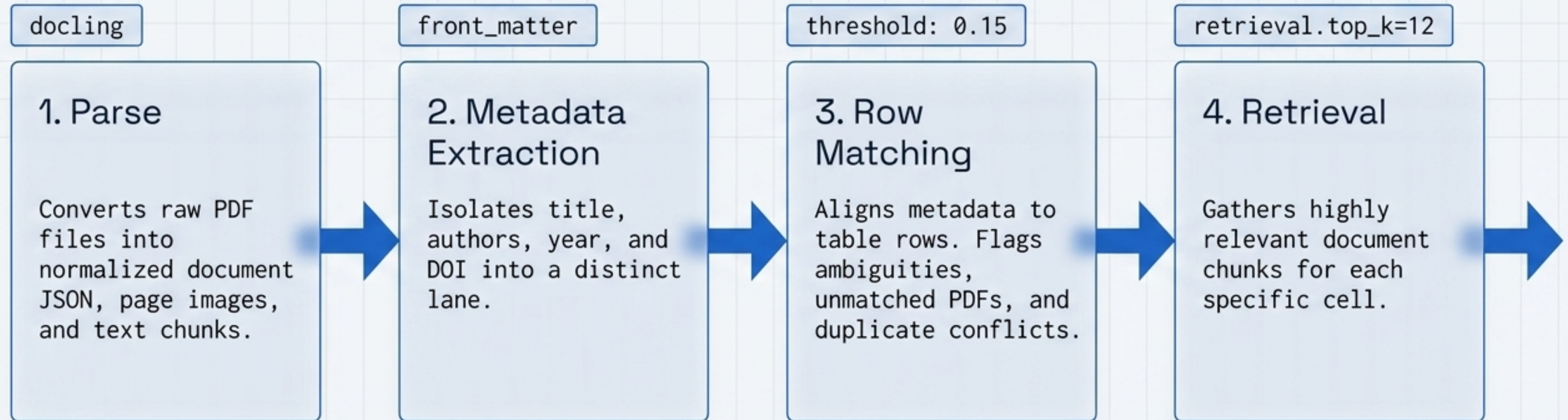


Evidence Ranking Rule — Source authority and field relevance always dictate the ranking order, never the raw model-return sequence.

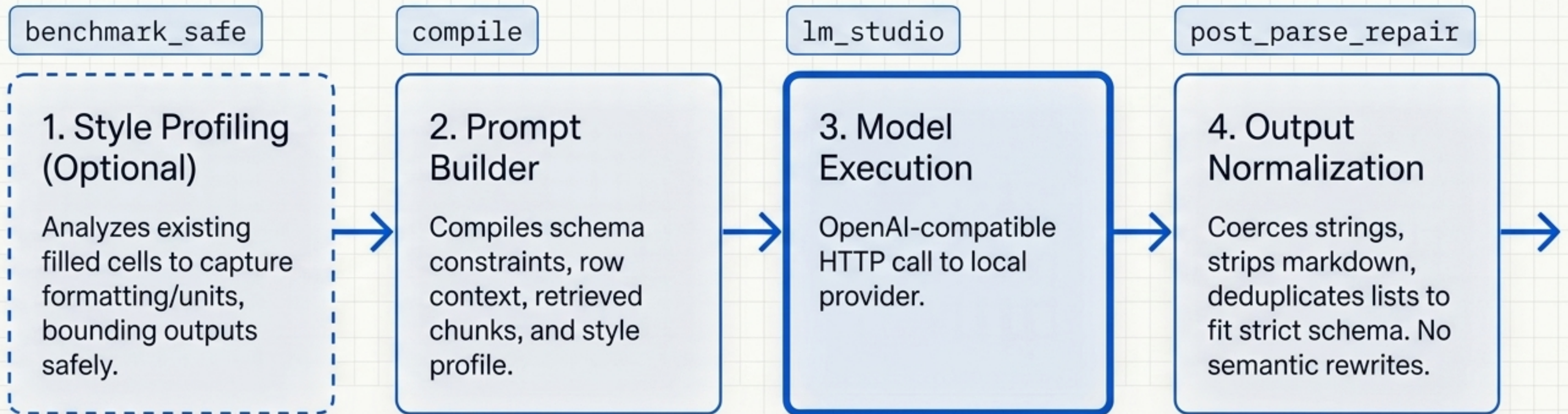
Three-Layered Architecture



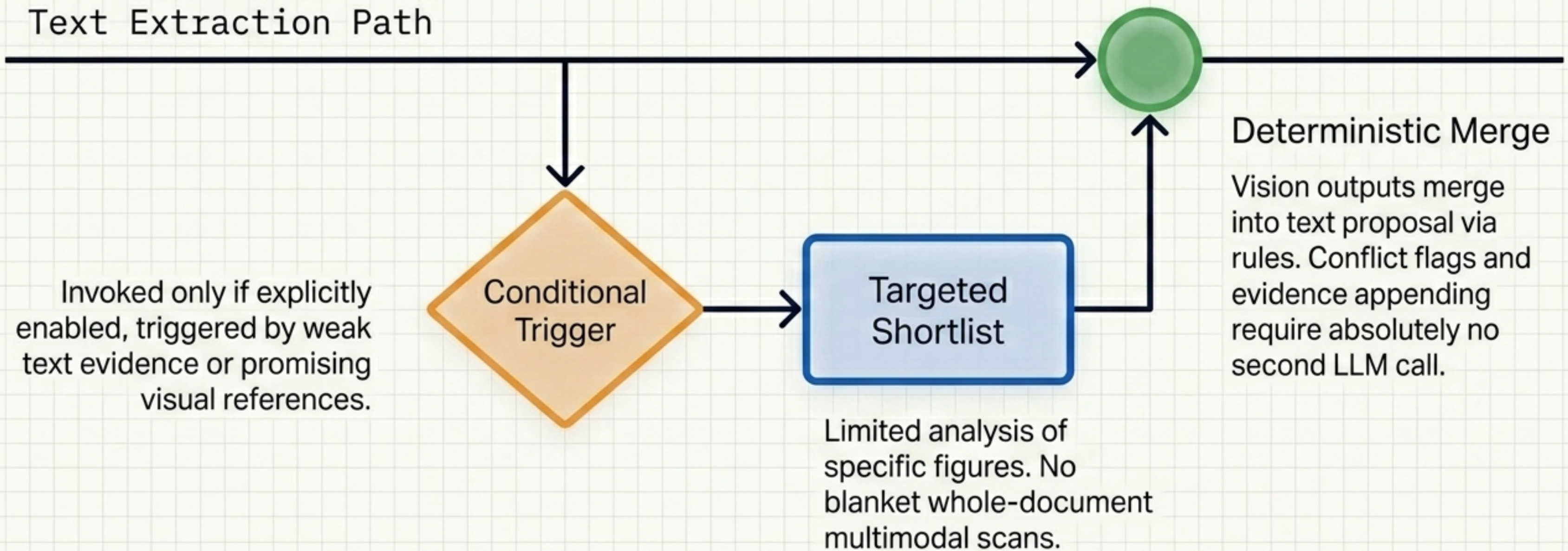
Extraction Pipeline: Preparation & Matching



Extraction Pipeline: Generation & Normalization



Deterministic Merging for Vision Extraction



Provider Readiness & Fallback Truth

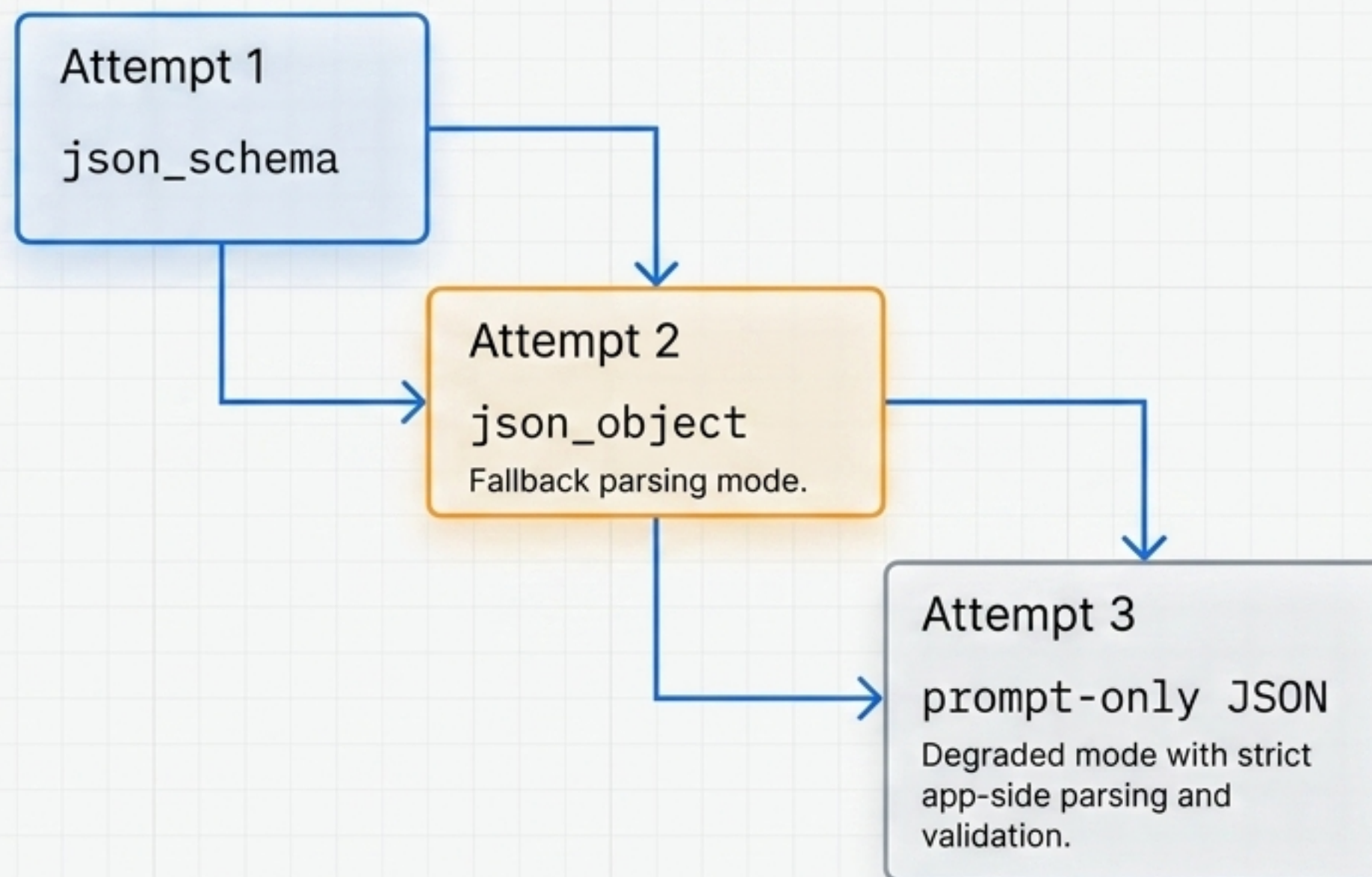
Connection Integrity

● LM Studio

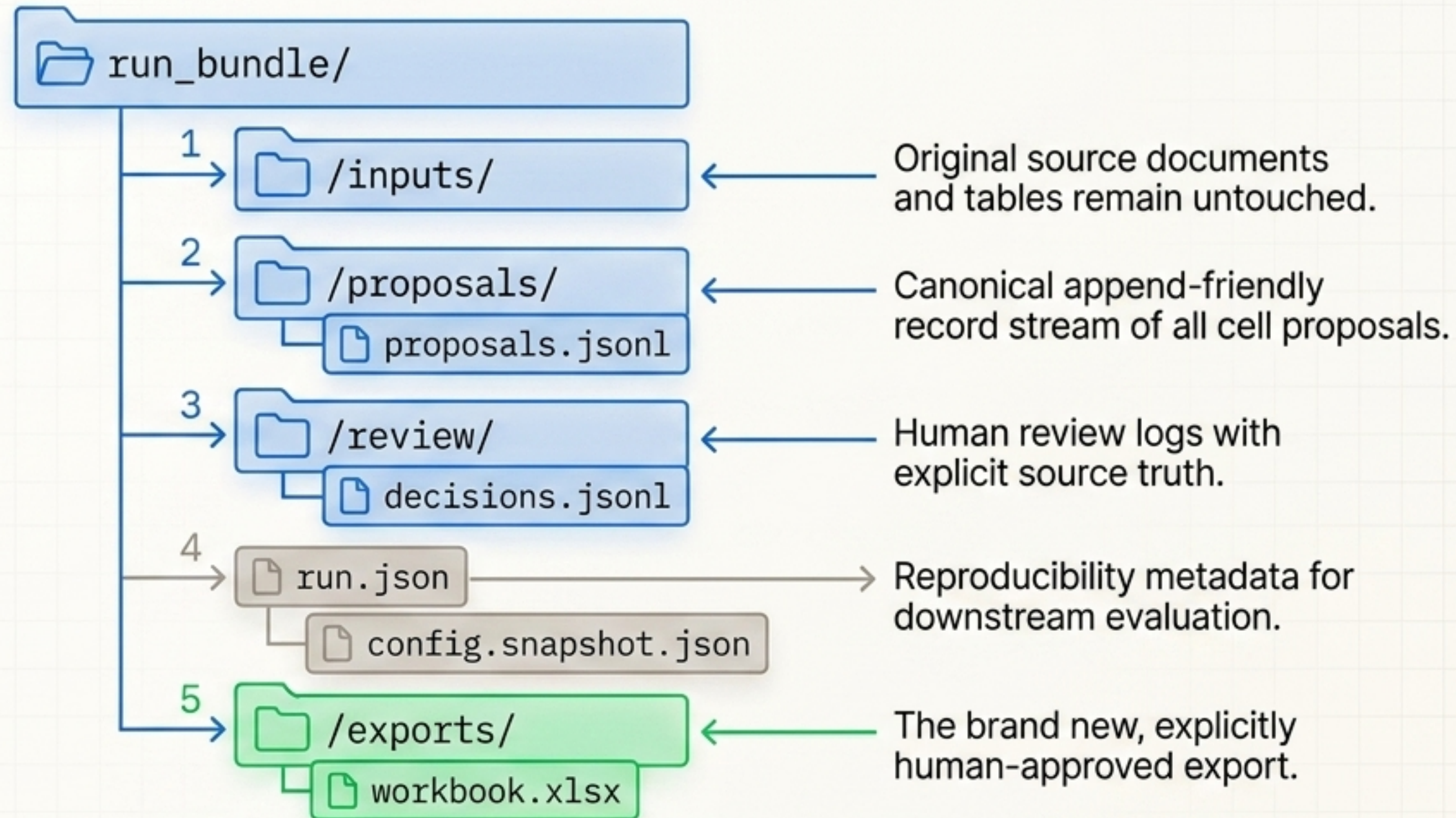
Strict Preflight: If lm_studio is unreachable at startup, run fails immediately. No cosmetically fake successful runs.

unsloth/gemma-4-26b-a4b-it

Structured Output Recovery Ladder



The Run Bundle Contract



Consumable from files alone. Downstream optimizer tools do not need to import main-app runtime code to interpret results.

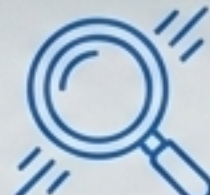
The Three Pillars of papers-to-table

Auditable Extraction



Schema-First Design

Driven entirely by strict target data definitions and bounded logic, explicitly rejecting open-chat interactions.



Evidence-Backed Truth

Every proposed value carries auditable, ranked source references, isolating exact quotes from inferred reasoning.



Human-Approved Export

Zero automated writes to the final workbook. Absolute authority rests with the human reviewer.